

Logic Operations Involving Binary Images

The language and theory of mathematical morphology often present a dual (but equivalent) view of binary images.

Thus, we have considered a binary image to be a bivalued function of spatial co-ordinates x and y .

Morphological theory views a binary image as a set of foreground (one valued) pixel and elements of which are in $\mathbb{Z}^2$ (Cartesian product) i.e., pair of elements.

Set operations such as union and intersection can be applied directly to binary image sets.

Example:

If A and B are binary images then $C = A \cup B$ is a binary image. where, a pixel in C is a foreground pixel if either or both of the corresponding pixel in A and B are foreground pixel.

The function C is given by.

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two view:

(i) First view

$$C(x, y) = \begin{cases} 1 & \text{if either } A(x, y) \text{ or } B(x, y) \text{ or both is } 1 \\ 0 & , \text{ otherwise} \end{cases}$$

(ii) Second view

$$C = \{(x, y) : (x, y) \in A \text{ or } (x, y) \in B \text{ or } (x, y) \in (A \text{ and } B)\}$$

where, A & B are binary image where the elements of A and B are 1-valued.

Thus, we see that the function point of view deals with the both foreground (1) and background (0) pixel simultaneously.

The set point of view deals only with foreground of pixel & it is understood that all pixels are not foreground pixel, that constitute the background.

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The set operations can be performed on binary images using following logical operations:

<u>Set Operation</u>	<u>Logical Operation</u>	<u>Name</u>
$A \cup B$	$A \vee B$	OR
$A \cap B$	$A \wedge B$	AND
A^c	$\sim A$	NOT
$A - B$	$A \wedge \sim B$	DIFFERENCE

V.Imp # Dilation & Erosion

Dilation :

Dilation is an operation that grows or thickens objects in an image. The specific manner & extent of this thickening is controlled by a set referred to as a structuring element.

Graphically structuring elements can be represented either by a matrix of 0's and 1's or as a set of foreground (1-valued) pixels.

So, the origin of the structuring element must be clearly identified by using the both representation.

(1)

Dilation operation can be done by using following steps:

(i) Representation of original image with rectangular object.

Let,

```

0 0 0 0 0 0 0 0 0 0 0
0 0 1 1 1 1 1 1 1 0 0
0 0 1 1 1 1 1 1 1 0 0
0 0 1 1 1 1 1 1 1 0 0
0 0 0 0 0 0 0 0 0 0 0
  
```

(ii) Structuring element with 5 pixels arranged in a diagonal line.

```

      1
     1
    1
   1
  1
 [1] ← origin or center of the
        structuring element as
        shown with a dark border
  
```

(iii) Structuring element translated to several location to the image

```

      * * * * *
    * * * * *
  1 1 1 1 1 1 1 * *
* 1 1 1 1 1 1 1 *
    * * 1 1 1 1 1 1
  * * * * *
* * * * *
  
```

does not overlap
1-valued pixel in
original image

overlap
1-valued pixel
in original
image

(5)

(iv) Output image ie shaded region will be replaced by 1's in the original image

```

      1 1 1 1 1 1 1 1 1
      1 1 1 1 1 1 1 1 1
      1 1 1 1 1 1 1 1 1
      1 1 1 1 1 1 1 1 1
      1 1 1 1 1 1 1 1 1
      1 1 1 1 1 1 1 1 1
      1 1 1 1 1 1 1 1 1
      1 1 1 1 1 1 1 1 1
      1 1 1 1 1 1 1 1 1
  
```

Now,

the dilation of A by B denoted $A \oplus B$ is defined as the set operation as:

$$A \oplus B = \{z / (\hat{B})_z \cap A \neq \emptyset\}$$

where,

$\emptyset$ = empty set

B = structuring element

The dilation of A by B is the set consisting of all the structuring element origin location where the reflected and translated B overlaps at least one element of A . It is a convention in image processing to the

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first operand of $A \oplus B$ be the image and second operand be the structuring element which usually is much smaller than the image.

Dilation is of two types as:

→ Dilation is associative:

$$A \oplus (B \oplus C) = (A \oplus B) \oplus C$$

→ Dilation is commutative:

$$A \oplus B = B \oplus A$$

Erosion:

Erosion is an operation that thins or shrinks objects in a binary image.

As in dilation, the manner and extent of erosion is controlled by a structuring element.

The erosion process can be done by using following steps:

(i) Representation of original image with rectangular object

0	0	0	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	1	1	0	0	0
0	0	1	1	1	1	1	1	1	0	0	0
0	0	1	1	1	1	1	1	1	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0

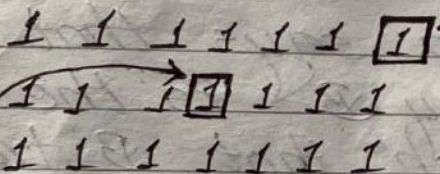
(7)

(i) Structuring element with 3-pixels arranged in vertical line



← The origin of the structuring element is shown with dark border

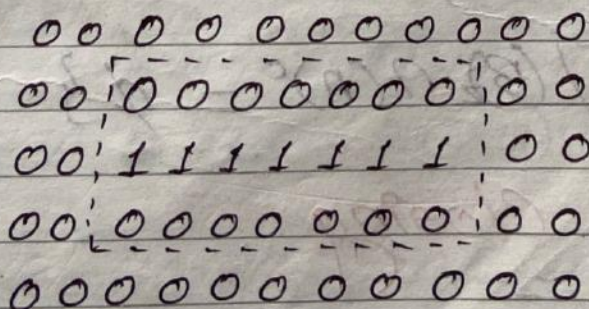
(ii) Structuring element translated into several location in the image



← The result is 0 at this location in the o/p image because all or part of the structuring element overlaps the background

The result is 1 at this location in the o/p image because the ~~structuring~~ structuring element fits entirely within the foreground.

(iv) Output image. The shaded region shows the location of 1's in the original image.



⑧

Now,
the erosion of A by B is denoted as $A \ominus B$ and is defined as:

$$A \ominus B = \{z \mid (B)_z \subseteq A\}$$

where,

$(B)_z \subseteq A$ means that $(B)_z$ is subset of A .

This equation says that the erosion of A by B is the set of all points z such that B translated by z is contained in A .

Thus, B is contained in A is equivalent to B not sharing any elements with the background of A .

The equivalent expression as the definition of erosion is defined as:

$$A \ominus B = \{z \mid (B)_z \cap (A)^c = \emptyset\}$$

Opening & Closing

Opening:

The morphological opening of A by B , denoted as $A \circ B$ is defined

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as the erosion of A by B and followed by a dilation of the result by B .

$$\text{i.e. } A \circ B = (A \ominus B) \oplus B$$

Now, an equivalent formulation of opening is:

$$A \circ B = \bigcup \{ (B)_z \mid (B)_z \subseteq A \}$$

Where,

$\bigcup \{ \}$ denotes the union of all sets inside the braces

$A \circ B$ is union of all translation of B that fits entirely within A .

Closing:

The morphological closing of A by B , denoted as $A \bullet B$ is a dilation of A by B and followed by an erosion of the result by B .

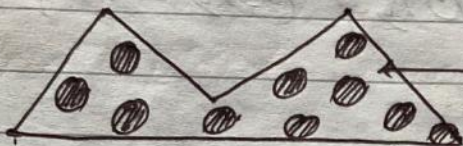
$$\text{i.e. } A \bullet B = (A \oplus B) \ominus B$$

Geometrically $A \bullet B$ is the complement of the union of all the translation of B that do not overlap A .

The opening and closing can be illustrated as:



fig (a): Set A & structuring element B



Translation of B inside A

fig (b): Translation of B that fit entirely within set A

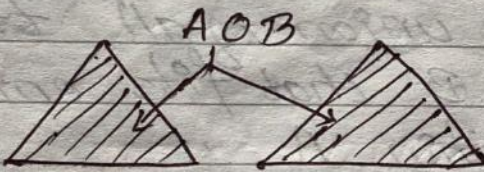


fig (c): The complete opening

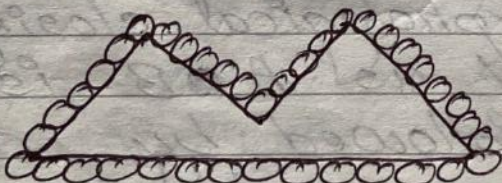


fig (d): Translation of B outside the border of A

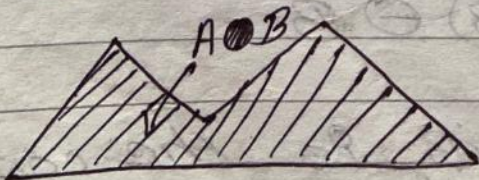


fig (e): The complete closing

(fig: Opening & Closing as Union of Translated Structuring Element)